Year 12 Further Maths - Regression

Similar questions - Least squares, coding and reliability

Two new questions on this topic, a little more open-ended than the exam questions. They are not past exam questions. Answers are at the end.

1 A biologist keeps six tanks at set water temperatures, $x\,^{\circ}\mathrm{C}$, and measures the mean mass, y kg, of the trout reared in each tank after a year. The results are given in the table.

x	12	14	15	17	18	21
y	2.9	2.6	2.7	2.2	2.0	1.6

(a) State which, if either, of x and y is an independent variable. **[1]**

(b) Calculate the equation of the least squares regression line of y on x. You are given $\sum x = 97$, $\sum y = 14$, $\sum x^2 = 1619$, $\sum xy = 218.7$. **[3]**

(c) State what effect, if any, there would be on your answer to part (b) if the masses had been recorded in grams rather than kilograms. **[2]**

(d) Explain what is meant by 'least squares', illustrating your answer with a sketch of a regression line, a data point and the quantity that is minimised. **[3]**

2

(a) Explain what is meant by 'least squares' in the context of a regression line of y on x. **[2]**

(b) A set of bivariate data (t,u) is summarised as follows.

$$n = 6, \quad \sum t = 30, \quad \sum u = 51, \quad \sum t^2 = 200, \quad \sum u^2 = 523, \quad \sum tu = 316$$

(i) Calculate the equation of the regression line of u on t. **[3]**

(ii) The variables t and u are now scaled using $v = 3t$ and $w = u - 2$. Find the equation of the regression line of w on v, giving your equation in the form $w = \mathrm{f}(v)$. **[2]**

3 A geography student measured the speed of flow, $y\ \mathrm{m\,s^{-1}}$, of water in a stream at points a distance x m from the bank, with $x = 1, 2, \ldots, 9$. The results are summarised by

$$n = 9, \quad \sum x = 45, \quad \sum y = 7.7, \quad \sum x^2 = 285, \quad \sum y^2 = 7.77, \quad \sum xy = 44.$$

(a) Explain why x is an example of an independent, controlled variable. **[1]**

A statistician believes that the point $(9, 0.6)$ may be an anomaly. For the other eight points, $\sum x = 36$, $\sum y = 7.1$, $\sum x^2 = 204$ and $\sum xy = 38.6$.

(b) Calculate the equation of the least squares regression line of y on x for the eight points, apart from $(9, 0.6)$. **[3]**

(c) Use your line to estimate the speed of flow when $x = 9$. **[1]**

(d) Use your answer to comment on the statistician's belief, and say why the estimate in part (c) must be treated with some caution. **[2]**

4 An insurance company collected data about the age, x years, of policyholders and the size of a claim, £y hundred. The data for 24 policyholders are summarised by

$$n = 24, \quad \sum x = 1008, \quad \sum y = 432, \quad \sum x^2 = 45300, \quad \sum y^2 = 9299, \quad \sum xy = 19100.$$

(a) Find the variance of x. **[1]**

(b) Find the equation of the regression line of y on x. **[2]**

(c) Hence estimate the expected size of claim from a policyholder of age 70. **[1]**

Tara is aged 70. She claims that the range of the data probably does not include people of her age because the mean age for the data is 42 and 70 is not close to this.

(d) Use your answer to part (a) to determine how likely it is that Tara's claim is correct. **[2]**

(e) Comment on the reliability of your estimate in part (c). You should refer to the value of the product-moment correlation coefficient for the data, which is 0.450 correct to 3 significant figures. **[2]**

5 The ages, x years, and the reaction times, t seconds, of a sample of 14 volunteers are summarised as follows.

$$n = 14, \quad \sum x = 700, \quad \sum t = 7.7, \quad \sum x^2 = 38000, \quad \sum t^2 = 4.275, \quad \sum xt = 391$$

(a) Calculate the value of the product-moment correlation coefficient between x and t. **[2]**

(b) Calculate the equation of the line of regression of t on x in the form $t = a + bx$. **[2]**

(c) Explain the relevance of the quantity $\sum(t - a - bx)^2$ to your answer to part (b). **[1]**

(d) Estimate the reaction time, in seconds, of a volunteer aged 42. **[1]**

It is subsequently decided to record the reaction time in tenths of a second.

(e) State what effect, if any, this change would have on (i) your answer to part (a), (ii) your answer to part (b). **[2]**

(f) It is known that the sample consisted almost entirely of students and retired people. Using this information and the value of the correlation coefficient found in part (a), comment on the reliability of your estimate in part (d). **[3]**

Question 1

(a) x (temperature) is the independent variable: it is controlled by the biologist, while y is measured

(b) $S_{xx} = 1619 - \dfrac{97^2}{6} = 50.83$; $S_{xy} = 218.7 - \dfrac{97 \times 14}{6} = -7.633$. $b = -0.150$; $a = \dfrac{14}{6} - b\dfrac{97}{6} = 4.76$. $y = 4.76 - 0.150x$

(c) Both a and b would be multiplied by 1000: $y = 4761 - 150x$ (approximately)

(d) The regression line minimises the sum of the squares of the vertical distances (residuals) from each data point to the line. (Sketch: line, a point, and a vertical segment from the point to the line labelled as the residual.)

Question 2

(a) The line is chosen to minimise $\sum(y_i - a - bx_i)^2$, the sum of the squares of the vertical distances (residuals) from the points to the line

(b)(i) $S_{tt} = 200 - \dfrac{30^2}{6} = 50$; $S_{tu} = 316 - \dfrac{30 \times 51}{6} = 61$. $b = 1.22$, $a = 8.5 - 1.22(5) = 2.4$. $u = 2.4 + 1.22t$

(ii) $w = u - 2 = 0.4 + 1.22t = 0.4 + 1.22\dfrac{v}{3}$, so $w = 0.4 + 0.407v$

Question 3

(a) x is set by the student (chosen in advance, with no error), whereas y is measured

(b) $S_{xx} = 204 - \dfrac{36^2}{8} = 42$; $S_{xy} = 38.6 - \dfrac{36 \times 7.1}{8} = 6.65$. $b = \dfrac{6.65}{42} = 0.158$; $a = \dfrac{7.1}{8} - 0.1583(4.5) = 0.175$. $y = 0.175 + 0.158x$

(c) $y = 0.175 + 0.1583(9) = 1.60\ \mathrm{m\,s^{-1}}$

(d) The measured value 0.6 is $1.0\ \mathrm{m\,s^{-1}}$ below the predicted value, supporting the belief that it is an anomaly. But $x = 9$ lies just outside the range of the data used for the line (extrapolation), so the estimate is less reliable

Question 4

(a) $\dfrac{45300}{24} - 42^2 = 123.5$

(b) $S_{xx} = 45300 - \dfrac{1008^2}{24} = 2964$; $S_{xy} = 19100 - \dfrac{1008 \times 432}{24} = 956$. $b = 0.3225$; $a = 18 - 0.3225(42) = 4.45$. $y = 4.45 + 0.3225x$

(c) $y = 4.45 + 0.3225(70) = 27.0$ (hundred pounds), i.e. about £2700

(d) Standard deviation $= \sqrt{123.5} = 11.1$. 70 is $\dfrac{70 - 42}{11.1} = 2.5$ standard deviations above the mean, so a policyholder of that age is unlikely to be included in the data range: Tara's claim is quite likely to be correct

(e) The estimate is an extrapolation (outside the range of the data), and $r = 0.450$ shows only a weak positive correlation, so the estimate is unreliable

Question 5

(a) $S_{xx} = 38000 - \dfrac{700^2}{14} = 3000$; $S_{tt} = 4.275 - \dfrac{7.7^2}{14} = 0.04$; $S_{xt} = 391 - \dfrac{700 \times 7.7}{14} = 6$. $r = \dfrac{6}{\sqrt{3000 \times 0.04}} = 0.548$

(b) $b = \dfrac{6}{3000} = 0.002$; $a = 0.55 - 0.002(50) = 0.45$. $t = 0.45 + 0.002x$

(c) It is the sum of the squares of the residuals, which the line of regression minimises

(d) $t = 0.45 + 0.002(42) = 0.534$ s

(e)(i) No effect (a linear scaling). (ii) a and b are both multiplied by 10: $t = 4.5 + 0.02x$

(f) A sample of mostly students and retired people means very few ages in the range 30–60, so 42 is in a sparsely sampled region. With $r = 0.548$ (moderate positive correlation) the estimate should be treated with caution